

FIITJEE – COMMON PHASE TEST

(JEE MAIN - 2024)
BATCHES – CTY 2224 X LOT (XII)
PHASE TEST - 4
CONDUCTION DATE: 12/08/2023

Q.P. Code:

Time Allotted: 3 Hours

Maximum Marks: 300

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

Important Instructions

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & C** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices ext. except the Admit Card inside the examination hall / room.

B. Filling of OMR Sheet:

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.
4. **Do not fold or make any stray marks on the Answer Sheet.**

C. Marking Scheme for All Two Parts:

- (i) **Part-A (01-20)** – Contains Twenty (20) multiple choice objective questions which have four (4) options each and only one correct option. Each question carries **+4 marks** which will be awarded for every correct answer and **-1 mark** will be deducted for every incorrect answer.
- (ii) **Part-C (01-05)** contains five (05) Numerical based questions with single digit integer as answer, ranging from 0 to 9 (both inclusive) and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

PHASE TEST - 4

Useful Data Chemistry:

Gas Constant	R	=	8.314 J K ⁻¹ mol ⁻¹
		=	0.0821 Lit atm K ⁻¹ mol ⁻¹
		=	1.987 $\approx$ 2 Cal K ⁻¹ mol ⁻¹
Avogadro's Number	N _a	=	6.023 $\times$ 10 ²³
Planck's Constant	h	=	6.626 $\times$ 10 ⁻³⁴ Js
		=	6.25 $\times$ 10 ⁻²⁷ erg.s
1 Faraday		=	96500 Coulomb
1 calorie		=	4.2 Joule
1 amu		=	1.66 $\times$ 10 ⁻²⁷ kg
1 eV		=	1.6 $\times$ 10 ⁻¹⁹ J
Atomic No :	H=1, D=1, Li=3, Na=11, K=19, Rb=37, Cs=55, F=9, Ca=20, He=2, O=8, Au=79.		
Atomic Masses:	He=4, Mg=24, C=12, O=16, N=14, P=31, Br=80, Cu=63.5, Fe=56, Mn=55, Si = 28 Pb=207, u=197, Ag=108, F=19, H=2, Cl=35.5		

Useful Data Physics:

Acceleration due to gravity $g = 10 \text{ m/s}^2$

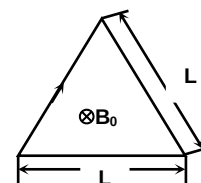
Section – I (Physics)

PART – A

Single Correct Choice Type

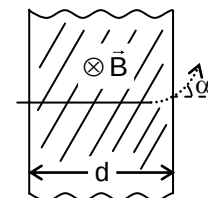
1. If a triangular loop (equilateral triangle) carrying a current I and uniform magnetic field B_0 is switched ON perpendicular inwards to the plane of the loop, then the net magnetic force acting on the loop is

(A) $3ILB_0$ (B) $3\sqrt{3}/2ILB_0$
(C) ILB_0 (D) zero



2. A proton (mass m and charge q) accelerated by a potential difference V flies through a uniform transverse magnetic field B . The field occupies a region of space of width ' d '. If ' α ' be the angle of deviation of proton from initial direction of motion (see figure), the value of $\sin \alpha$ will be

(A) $\frac{B}{2} \sqrt{\frac{qd}{mV}}$ (B) $qV \sqrt{\frac{Bd}{2m}}$
(C) $\frac{B}{2} \sqrt{\frac{q}{2mV}}$ (D) $Bd \sqrt{\frac{q}{2mV}}$



3. When a charged particle moves perpendicular to a magnetic field, then :

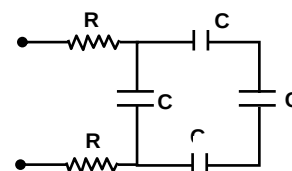
(A) Speed of the particle is changed
(B) Speed of the particle remains unchanged
(C) Direction of the particle remains unchanged
(D) Acceleration of the particle remains unchanged

4. A cell of e.m.f. E and internal resistance r is connected in series with an external resistance nr then the ratio of the terminal potential difference to E.M.F. is

(A) $1/n$ (B) $\frac{1}{n+1}$ (C) $\frac{n}{n+1}$ (D) $\frac{n+1}{n}$

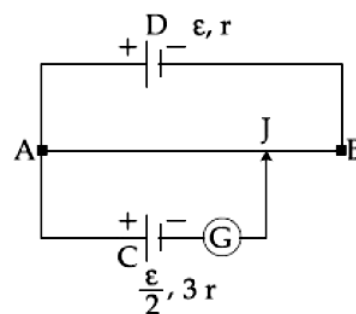
5. The time constant of the circuit shown in the figure is

(A) $\frac{RC}{3}$ (B) $\frac{4RC}{3}$
(C) $\frac{2RC}{3}$ (D) $\frac{8RC}{3}$



6. A potentiometer wire AB having length L and resistance $12r$ is joined to a cell D of emf ϵ and internal resistance r . A cell C having emf $\epsilon/2$ and internal resistance $3r$ is connected. The length AJ at which the galvanometer as shown in figure shows no deflection is:

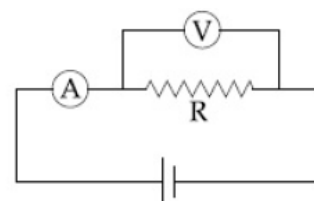
(A) $\frac{11}{12}L$ (B) $\frac{11}{24}L$
(C) $\frac{13}{24}L$ (D) $\frac{5}{12}L$



7. The actual value of resistance R , shown in the figure is 30Ω . This is measured in an experiment as shown using the standard formula $R = \frac{V}{I}$, where V and I are the readings of the voltmeter and ammeter, respectively.

If the measured value of R is 5% less, then the internal resistance of the voltmeter is :

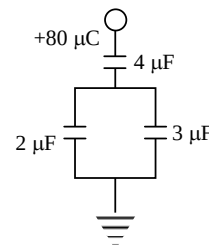
(A) 600Ω (B) 570Ω
(C) 35Ω (D) 350Ω



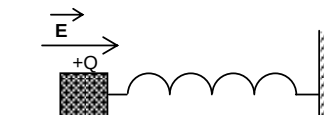
8. In the given circuit, a charge of $+80 \mu\text{C}$ is given to the upper plate of the $4 \mu\text{F}$ capacitor. Then in the steady state, the charge on the upper plate of the $3 \mu\text{F}$ capacitor is

(A) $+32 \mu\text{C}$
(C) $+48 \mu\text{C}$

(B) $+40 \mu\text{C}$
(D) $+80 \mu\text{C}$



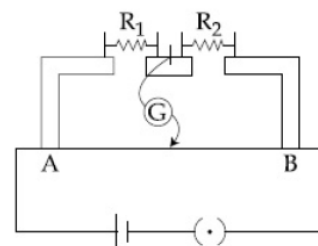
9. A wooden block performs SHM on a frictionless surface with frequency, ν_0 . The block carries a charge $+Q$ on its surface. If now a uniform electric field $\vec{E}$ is switched-on as shown, then the SHM of the block will be
- (A) of the same frequency and with shifted mean position.
(B) of the same frequency and with the same mean position
(C) of changed frequency and with shifted mean position.
(D) of changed frequency and with the same mean position.



10. In the experimental setup of metre bridge shown in the figure, the null point is obtained at a distance of 40cm from A. If a 10Ω resistor is connected in series with R_1 , the null point shifts by 10cm. The resistance that should be connected in parallel with $(R_1 + 10)\Omega$ such that the null point shifts back to its initial position is:

(A) 20Ω
(C) 60Ω

(B) 40Ω
(D) 30Ω



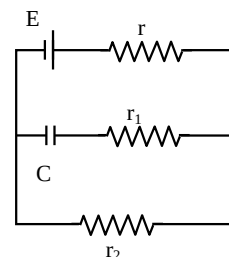
11. In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance C will be:

(A) $\frac{1}{9}$

(B) CE

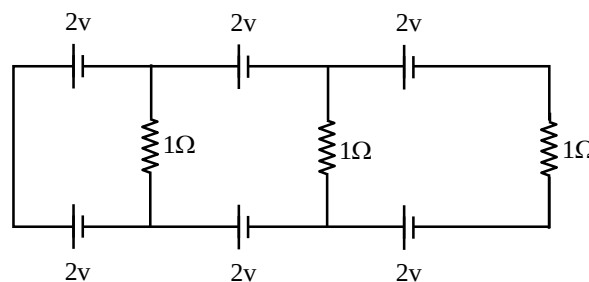
(C) $CE \frac{r_1}{(r_2 + r)}$

(D) $CE \frac{r_2}{(r + r_2)}$



12. In the given circuit, the current in each resistance is:

(A) 0 A
(B) 1 A
(C) 0.25 A
(D) 0.5 A



13. A long straight wire along the z -axis carries a current I in the negative z -direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the $z = 0$ plane is

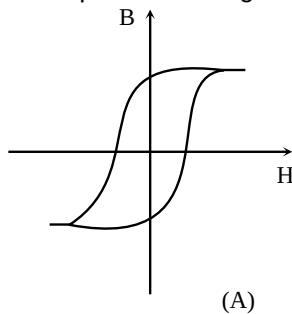
(A) $\frac{\mu_0 I}{2\pi} \frac{(y\hat{i} - x\hat{j})}{(x^2 + y^2)}$

(B) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{i} + y\hat{j})}{(x^2 + y^2)}$

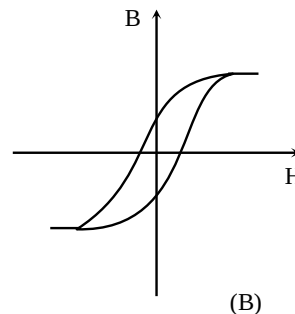
(C) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{j} - y\hat{i})}{(x^2 + y^2)}$

(D) $\frac{\mu_0 I}{2\pi} \frac{(x\hat{i} - y\hat{j})}{(x^2 + y^2)}$

14. Hysteresis loops for two magnetic materials A and B are given below :



(A)



(B)

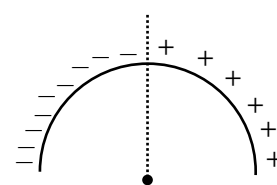
These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use :

- (A) A for electromagnets and B for electric generators
 (B) A for transformers and B for electric generators
 (C) B for electromagnets and transformers
 (D) A for electric generators and transformers
15. A rigid circular loop of radius r and mass m lies in the x - y plane on a flat table and has a current I flowing in it. At this particular place. The earth's magnetic field is $\vec{B} = B_x \hat{i} + B_y \hat{j}$. The minimum value of I for which one end of the loop will lift from the table is

- (A) $\frac{mg}{\pi r B_x}$ (B) $\frac{mg}{\pi B_y}$ (C) $\frac{mg}{2\pi r \sqrt{B_x^2 + B_y^2}}$ (D) $\frac{mg}{\pi r \sqrt{B_x^2 + B_y^2}}$

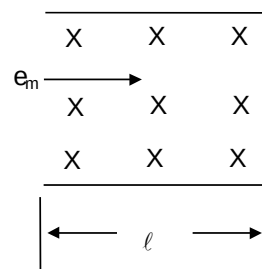
16. Half part of semicircular ring of radius R is positively charged and other half is negatively charged. The dipole moment of the arrangement is (magnitude of charge density $= \lambda$)

- (A) $2 \lambda R^2$ (B) $4 \lambda R^2$
 (C) $\pi R^2 \lambda$ (D) $2 \pi R^2 \lambda$

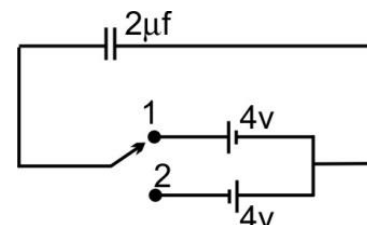


17. An electron moves straight inside a charged parallel plate capacitor of uniform surface charge density σ . The space between the plates is filled with constant magnetic field of induction $\vec{B}$. The time of straight line motion of the electron in the capacitor is

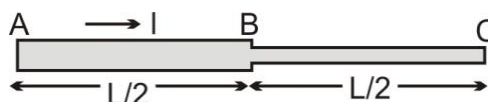
- (A) $\frac{e\sigma}{\epsilon_0 \ell B}$ (B) $\frac{\epsilon_0 \ell B}{\sigma}$
 (C) $\frac{e\sigma}{\epsilon_0 B}$ (D) $\frac{\epsilon_0 B}{e\sigma}$



18. Heat generated in the circuit when switch is shifted from 1 to 2 is
 (A) 32 μ J (B) 16 μ J
 (C) 48 μ J (D) 64 μ J



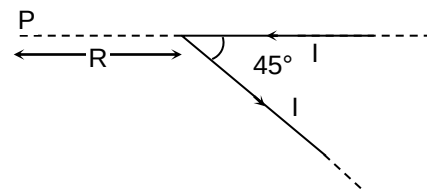
19. Two wires AB and BC, each of length $L/2$ are made of the same material. The radius of wire AB is $2r$ and of wire BC is r . The current I flows through the composite wire. Choose the correct statement from the following



- (A) Potential difference across BC is twice that across AB
 (B) Power dissipated in BC is four times the power dissipated in AB
 (C) Current densities in AB and BC are equal

(D) Electric fields in AB and BC are equal

20. A long straight wire, carrying current I , is bent at its midpoint to form an angle of 45° . Magnetic field at point P, distance R from point of bending is equal to

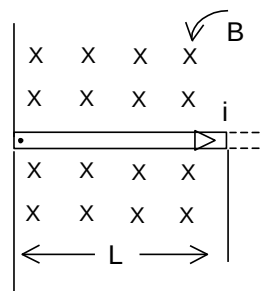


- (A) $\frac{(\sqrt{2}-1)\mu_0 I}{4\pi R}$ (B) $\frac{(\sqrt{2}+1)\mu_0 I}{4\pi R}$
 (C) $\frac{(\sqrt{2}+1)\mu_0 I}{4\sqrt{2}\pi R}$ (D) $\frac{(\sqrt{2}-1)\mu_0 I}{2\sqrt{2}\pi R}$

PART - C Integer Type

1. A dipole of moment $\vec{p} = 10^{-7}(5\hat{i} + \hat{j} - 2\hat{k})$ coulomb meter is placed in an electric field $\vec{E} = 10^7(\hat{i} + \hat{j} + \hat{k})$ newton/coulomb. The magnitude of the torque (Nm) experienced by dipole is $k\sqrt{18.5}$. Find the value of k .
2. A 100 pF capacitor is charged to a potential difference of 24 V. It is connected to an uncharged capacitor of 20 pF. The new potential difference across the 100 pF capacitor is K volt. Find the value of K/4.

3. A straight conductor of mass m and carrying a current i is hinged at one end and placed in a plane perpendicular to the magnetic field of intensity $\vec{B}$ as shown in the figure. At any moment if the conductor is let free, the angular acceleration of the conductor will be $\frac{N}{2}$. The value of N is [neglect gravity, use $iB = m$ (numerically)].

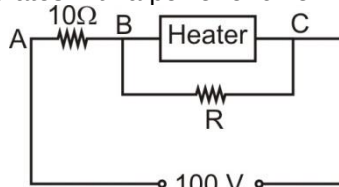


4. For spherical charge distribution given as

$$\begin{cases} \rho = \rho_0 \left(1 - \frac{r}{3}\right) & , \quad \text{when } r \leq 3 \\ \rho = 0 & , \quad \text{when } r > 3 \end{cases}$$

(where r is the distance from the centre of spherical charge distribution)
 The electric field intensity is maximum for what value of r ?

5. A heater is designed to operate with a power of 1000W in a 100 V line. It is connected, in combination with a resistance of 10Ω and a resistance R to a 100 V line as shown in figure. What should be the value of R so that the heater operates with a power of 62.5 W ?



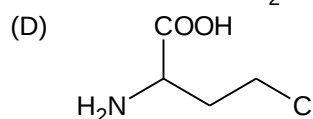
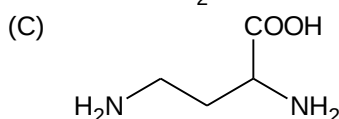
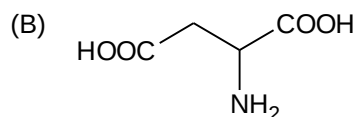
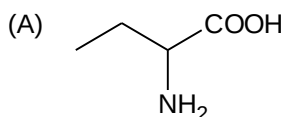
pace for rough work

Section – II (Chemistry)

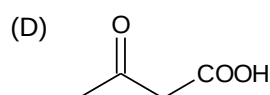
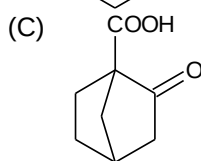
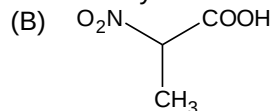
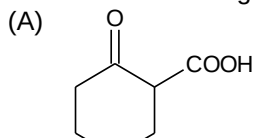
PART – A

Single Correct Choice Type

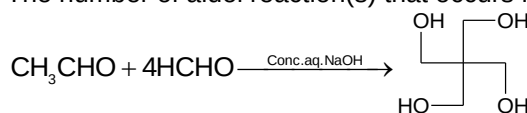
- What is the degree of unsaturation of the compound formed by the reaction of formaldehyde with excess of ammonia?
(A) 6 (B) 4 (C) 3 (D) 5
- D-glucose and D-fructose can be distinguished by
(A) Fehling's Test (B) Tollen's Test (C) $\text{Br}_2/\text{H}_2\text{O}$ (D) Osazone formation
- Which of the amino acids is a basic amino acid?



- Which of the following compounds is most difficult to decarboxylate?

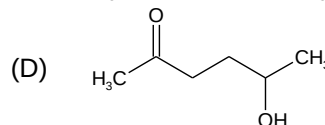
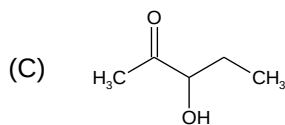
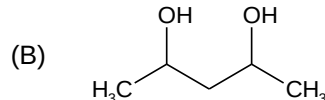
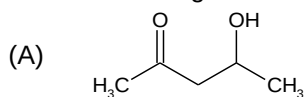


- Phenol does not react with NaHCO_3 because
(A) phenol is a weaker acid than carbonic acid (B) phenol is a stronger acid than carbonic acid
(C) phenol is as strong as carbonic acid (D) phenol is insoluble in water
- The number of aldol reaction(s) that occurs in the given transformation is



- (A) 1 (B) 2 (C) 3 (D) 4
- The strongest acid among the following aromatic compounds is
(A) p-chlorophenol (B) p-nitrophenol (C) m-nitrophenol (D) o-nitrophenol
- $\text{B(mix)} \xleftarrow[2]{\text{conc. HI}} (\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3 \xrightarrow[1]{\text{anhydrous HI}} \text{A (mix)}$
(A) A and B are identical mixture of CH_3I and $(\text{CH}_3)_3\text{C}-\text{OH}$
(B) A and B are identical mixture of CH_3OH and $(\text{CH}_3)_3\text{C}-\text{I}$
(C) A is mixture of CH_3I and $(\text{CH}_3)_3\text{C}-\text{OH}$ and B is a mixture of CH_3OH and $(\text{CH}_3)_3\text{C}-\text{I}$
(D) none of the above
- A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives
(A) benzyl alcohol and sodium formate (B) sodium benzoate and methyl alcohol
(C) sodium benzoate and sodium formate (D) None of the above

10. Which of the following will be most ready dehydrated in acidic conditions?

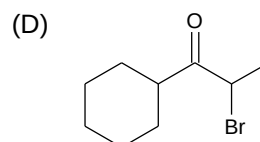
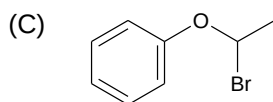
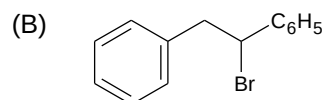
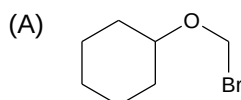


11. Aldol condensation will not be observed in

(A) chloral
(C) hexanal

(B) phenylacetaldehyde
(D) Acetone

12. Which of the following, upon treatment with tert-BuONa followed by addition of bromine water, fails to decolourise the colour of bromine?

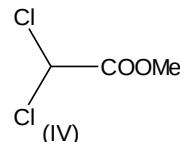
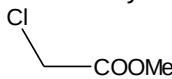
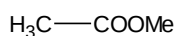
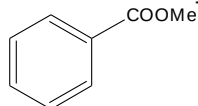


13. The most reactive compound towards formation of cyanohydrin on treatment with KCN followed by acidification is

(A) benzaldehyde
(C) phenylacetaldehyde

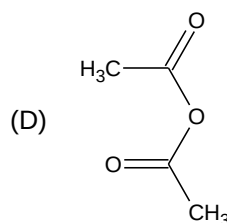
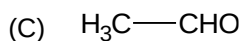
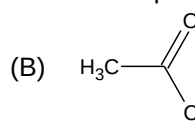
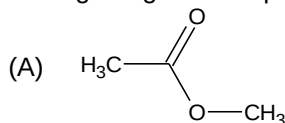
(B) p-nitrobenzaldehyde
(D) p-hydroxybenzaldehyde

14. The order of reactivity of the following ester towards hydrolysis is



- (A) I > II > III > IV (B) II > I > III > IV (C) IV > III > II > I (D) IV > III > I > II

15. Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is



16. When propanoic acid is treated with aqueous sodium bicarbonate, CO₂ is liberated. The C of CO₂ comes from

(A) carboxylic acid group
(C) bicarbonate

(B) methylene group
(D) methyl group

17. Nylon-66 is a polyamide of

(A) Vinylchloride and formaldehyde
(C) Adipic acid and hexamethylene diamine

(B) Adipic acid and methyl amine
(D) Formaldehyde and melamine

18. To separate a mixture of amines from each other one should follow

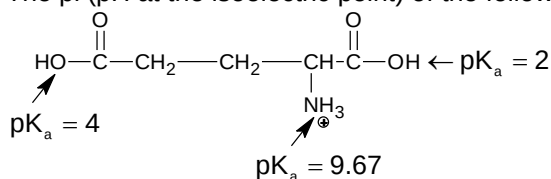
(A) Beckmann's method

(B) Hinsberg's method

19. (C) Zeisel method (D) Victor Meyer's Method
Among the following which will not respond to iodoform test
- (A) $\text{H}_3\text{C}-\text{CH}_2-\text{OH}$ (B) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_2-\text{C}(=\text{O})-\text{CH}_3$
- (C) $\text{H}_3\text{C}-\text{CH}(\text{OH})-\text{CH}_3$ (D) CCl_3CHO
20. Which of the following is not a condensation polymer?
(A) Glyptal (B) Nylon-66 (C) Dacron (D) PTFE(Teflon)

PART-C Integer Type

- If 'x' is the number of non-reducing sugars in the list mentioned as under, then the value of x is D-glucose, sucrose, Maltose, Lactose, Glycogen, Dextrin, Cellulose, D-fructose, amylopectin.
- What is the number of asymmetric carbon atoms in open chain structure of aldopentose?
- How many isomeric amines with formula $\text{C}_7\text{H}_9\text{N}$ contain a benzene ring?
- The number of moles of phenylhydrazine required to convert 1 mole of glucose to glucosazone is y. The value of y^2 is:
- The pI (pH at the isoelectric point) of the following amino acid:



pace for rough work

Section – III (Mathematics)

PART – A

Single Correct Choice Type

1. Domain of the function $y = \cos^{-1}\left(\frac{1+x^4}{2x^2}\right) + \sqrt{x(x+2)}$ is
 (A) $[-1, 1]$ (B) $(-1, 1)$ (C) $\{-1, 1\}$ (D) $\{1\}$
2. Which of the following is an even function, where $[]$ stands for the greatest integer function
 (A) $f(x) = \left\lfloor \frac{x}{2} \right\rfloor + \frac{1}{2}$ (B) $f(x) = [x - [x]]$
 (C) $f(x) = [\sin x + \cos x]$ (D) None of these
3. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}$ then for all x , $f(g(x))$ equals
 (A) x (B) 1 (C) $f(x)$ (D) $g(x)$
4. X and Y are two sets and $f : X \rightarrow Y$. If $\{f(c) = y; c \in X, y \in Y\}$ and $\{f^{-1}(d) = x; d \in Y, x \in X\}$, then the true statement is
 (A) $f(f^{-1}(b)) = b$ (B) $f^{-1}(f(a)) = a$
 (C) $f(f^{-1}(b)) = b, b \in y$ (D) $f^{-1}(f(a)) = a, a \in x$
5. $\int_{-2016}^{2016} \min(\{x\}, \{-x\}) dx$ is equal to (where $\{.\}$ denotes the fractional part)
 (A) 0 (B) 508 (C) 1008 (D) 2015
6. $\int_{1/4}^4 \frac{1}{x} \sin\left(x - \frac{1}{x}\right) dx =$
 (A) $-\sin\left(\frac{15}{4}\right)$ (B) $\sin\left(\frac{15}{4}\right)$ (C) $\frac{15}{4}$ (D) 0
7. Let f be a real valued function defined on $(-1, 1)$ such that $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt \forall x \in (-1, 1)$ and f^{-1} be the inverse of ' f ' then $(f^{-1})'(2)$ is
 (A) 1 (B) $1/2$ (C) $1/3$ (D) $1/e$
8. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r^3}{r^4 + n^4} \right) =$
 (A) $\log 2$ (B) $\frac{1}{2} \log 2$ (C) $\frac{1}{3} \log 2$ (D) $\frac{1}{4} \log 2$
9. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2 + 1} + \frac{2}{n^2 + 2} + \frac{3}{n^2 + 3} + \dots + \frac{1}{n+1} \right] =$
 (A) 0 (B) $1/2$ (C) 1 (D) 2
10. $\lim_{n \rightarrow \infty} \left(\frac{n^2 + n + 1}{(n-1)(n-2)} \right)^n$ equals
 (A) e^4 (B) e^{-4} (C) e^2 (D) e^{-2}

11. If the function $f(x) = \begin{cases} \frac{x^2}{a}, & 0 \leq x < 1 \\ a, & 1 \leq x < \sqrt{2} \\ \left(\frac{2b^2 - 4b}{x^2}\right), & \sqrt{2} \leq x < \infty \end{cases}$

is continuous for $0 \leq x < \infty$, then which of the following is true?

- (A) $a = 1, b = -1$ (B) $a = -1, b = 1 + \sqrt{2}$
(C) $a = -1, b = 1$ (D) $a = -1, b = -1$

12. Let $f(x) = \begin{cases} \frac{e^{\alpha x} - e^x - x}{x^2}, & x \neq 0 \\ \frac{3}{2}, & x = 0 \end{cases}$

then the value of α for which $f(x)$ is a continuous function at $x = 0$ will be

- (A) 1 (B) 0 (C) 4 (D) 2

13. The graph of $y = x^3 + ax^2 + bx + c$ has no extremum if

- (A) $a^2 < 3b$ (B) $a^2 > 3b$ (C) $b^2 - 4ac < 0$ (D) $b^2 - 4ac > 0$

14. If $f(x) = \begin{cases} \cos \frac{\pi x}{2015}, & x > 0 \\ 2x + a, & x \leq 0 \end{cases}$, then the values of a if $x = 0$ is a point of local maxima for $f(x)$.

- (A) $a \geq 0$ (B) $a \leq 0$ (C) $a \geq 1$ (D) $a \leq 1$

15. The curves $4x^2 + 9y^2 = 72$ and $x^2 - y^2 = 5$ at $(3, 2)$

- (A) touch each other (B) cut orthogonally
(C) intersect at 45° (D) intersect at 60°

16. If $6(8a + c) = 16b + 3d$, then $f(x) = ax^3 + bx^2 + cx + d$ has at least one root in

- (A) $(-3, 0)$ (B) $(-4, 0)$ (C) $(-4, -3)$ (D) $(0, 2)$

17. If $I = \int \frac{dx}{4 + 5 \sin x} = \frac{1}{p} \log \left| \frac{q \tan(x/2) + 1}{\tan(x/2) + r} \right| + C$ then $p - q + r =$

- (A) 2 (B) 3 (C) $\sqrt{3}$ (D) $4 + \sqrt{3}$

18. The value of $\int \frac{(x^4 - x)^{\frac{1}{4}}}{x^5} dx$ is equal to

- (A) $\frac{1}{15} \left(1 - \frac{1}{x^3}\right)^{\frac{5}{4}} + c$ (B) $\frac{4}{15} \left(1 - \frac{1}{x^3}\right)^{\frac{5}{4}} + c$ (C) $\frac{1}{4} \left(1 - \frac{1}{x^3}\right)^{\frac{5}{4}} + c$ (D) $\frac{3}{4} \left(1 - \frac{1}{x^3}\right)^{\frac{5}{4}} + c$

19. $\int \sqrt{\frac{\cos x - \cos^3 x}{1 - \cos^3 x}} dx$ is equal to

- (A) $\frac{2}{3} \sin^{-1}(\cos^{3/2} x) + C$ (B) $\frac{3}{2} \sin^{-1}(\cos^{3/2} x) + C$
(C) $\frac{2}{3} \cos^{-1}(\cos^{3/2} x) + C$ (D) none of these

20. If $y = e^{\sqrt{x}} + e^{-\sqrt{x}}$ then $x \frac{d^2 y}{dx^2} + (1/2) \frac{dy}{dx}$ is equal to
- (A) y (B) $x(e^{\sqrt{x}} + e^{-\sqrt{x}})$ (C) $(1/4)y$ (D) $\sqrt{x}y$

PART – C
Integer Type

1. Number of solution of equation $2^{\cos x} = |\sin x|$ in $[0, 2\pi]$ is....
2. Let $f_1(x) = \frac{x}{2} + 4 \quad \forall x \in \mathbb{R}$ and define $f_n(x) = f_1(f_{n-1}(x)) \quad \forall n \geq 2$ then $\lim_{n \rightarrow \infty} f_n(x)$ is ...
3. If $f(x) + f(x+1) + f(x+2) = 3x^2 + 2x + 1$, then $\int_0^3 f(x) dx$ is...
4. Total number of points of discontinuity in $(0, 3)$ of the function $f(x) = [3x] + \{2x\}$ (where $[.]$ denotes G.I.F and $\{.\}$ denotes fractional part) is/are.....
5. If $f(x) = \begin{cases} |x-1| + \lambda, & x \leq 1 \\ 2x+3, & x > 1 \end{cases}$ has a local minima at $x=1$, then maximum value of λ is....

space for rough work

FIITJEE COMMON TEST

BATCHES: CTY 2224 X LOT
CLASS- XII
PHASE TEST – 4 (12-08-2023)
QP. Code: 00000
JEE MAIN
ANSWER KEY

SECTION - I: PHYSICS

PART - A

1. D	2. D	3. B	4. C
5. D	6. C	7. B	8. C
9. A	10. C	11. D	12. A
13. A	14. C	15. D	16. A
17. B	18. D	19. B	20. A

PART - C

1. 2	2. 5	3. 3	4. 2
5. 5			

SECTION - II: CHEMISTRY

PART – A

1. C	2. C	3. C	4. C
5. A	6. C	7. B	8. C
9. A	10. A	11. A	12. A
13. B	14. C	15. B	16. C
17. C	18. B	19. D	20. D

PART - C

1. 5	2. 3	3. 5	4. 9
5. 3			

SECTION- III: MATHEMATICS

PART – A

1. D	2. B	3. B	4. D
5. C	6. D	7. C	8. D
9. B	10. A	11. C	12. D
13. A	14. C	15. B	16. B
17. B	18. B	19. C	20. C

PART – C

1. 4	2. 8	3. 3	4. 9
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5. 5



HINTS & SOLUTIONS

SECTION - I: PHYSICS

PART - A

1. D
2. D
3. B
4. C
5. D
6. C
- Sol. $V_{AJ} = IR_{AJ}$

$$\frac{E}{2} = \left(\frac{E}{12r} \right) \times \left(\frac{x}{L} \times 12r \right)$$
7. B
- Sol. $\frac{30R}{R+30} = 30 \times 0.95$
 $\Rightarrow R = 570 \Omega$
8. C
- Sol. Let 'q' be the final charge on $3\mu F$ capacitor then

$$\frac{80-q}{2} = \frac{q}{3} \Rightarrow q = 48\mu C$$
9. A
- Sol. Constant force does not change the frequency or T of SHM. It just changes the mean position.
10. C
- Sol. $\frac{R_1}{R_2} = \frac{2}{3} \dots(i)$

$$\frac{R_1+10}{R_2} = 1$$

 $\Rightarrow R_1 + 10 = R_2 \dots(ii)$

$$\frac{2R_2}{3} + 10 = R_2; 10 = \frac{R_2}{3}$$

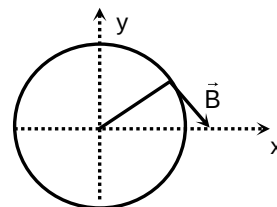
 $\Rightarrow R_2 = 30 \Omega \text{ \& } R_1 = 20 \Omega$

$$\frac{30 \times R}{30+R} = \frac{2}{3}$$

 $R = 60 \Omega$
11. D
- Sol. $q = CV = \frac{CEr_2}{r+r_2}$
12. A
- Sol. Potential difference across each resistor is zero.
13. A
- Sol. Magnetic field $|\vec{B}| = \frac{\mu_0 I}{2\pi\sqrt{x^2+y^2}};$
 Unit vector perpendicular to the position vector is

$$\frac{(y\hat{i} - x\hat{j})}{\sqrt{x^2+y^2}}$$

$$\therefore |\vec{B}| = \frac{\mu_0 I}{2\pi(x^2+y^2)}(y\hat{i} - x\hat{j})$$
14. C
- Sol. For electromagnet and transformer, the coercivity should be low to reduce energy loss
15. D
- Sol. $\vec{\gamma} = \vec{\mu} \times \vec{\beta}$

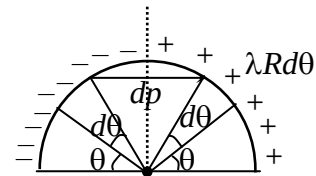


$$mgr = i\pi r^2 \sqrt{B_x^2 + B_y^2}$$

$$\therefore i = \frac{mg}{\pi r \sqrt{B_x^2 + B_y^2}}$$

16. A

$$\begin{aligned} \text{Sol. } \int d\vec{p} &= \int_0^{\pi/2} (\lambda R d\theta) 2R \cos\theta \\ &= 2\lambda R^2 \int_0^{\pi/2} \cos\theta d\theta = 2\lambda R^2 \end{aligned}$$



17. B

Sol. The net electric field

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

$$\Rightarrow \vec{E} = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

The net force acting on the electron is zero because it moves with constant velocity, due to its motion on straight line.

$$\Rightarrow \vec{F}_{\text{net}} = \vec{F}_e + \vec{F}_m = 0$$

$$\Rightarrow |\vec{F}_e| = |\vec{F}_m|$$

$$\Rightarrow eE = eVB \Rightarrow V = \frac{E}{B} = \frac{\sigma}{\epsilon_0 B}$$

$\therefore$ The time of motion inside the capacitor

$$= t = \frac{\ell}{v} = \frac{\epsilon_0 \ell B}{\sigma}$$

18. D

 Sol. Δu in capacitor = 0

Work by battery = $qE = 64 \mu\text{J}$

$\therefore$ Heat = $64\mu\text{J}$

19. B

Sol. Current flowing through both the bars is equal. Now, the heat produced is given by

$$\begin{aligned} H &= I^2 R t \text{ or } H \propto R \text{ or } \frac{H_{AB}}{H_{BC}} = \frac{R_{AB}}{R_{BC}} \\ &= \frac{(1/2r)^2}{(1/r)^2} \quad \left(\text{as } R \propto \frac{1}{A} \propto \frac{1}{r^2} \right) \\ &= \frac{1}{4} \text{ or } H_{BC} = 4H_{AB} \end{aligned}$$

20. A

PART - C

1. 2

2. 5

3. 3

Sol. The force acting on the elementary portion of the current carrying conductor is given as,
 $d\vec{F} = i(dr)B \sin 90^\circ$

$$\Rightarrow d\vec{F} = iBdr$$

The torque applied by $d\vec{F}$ about $O = d\tau = r dF$

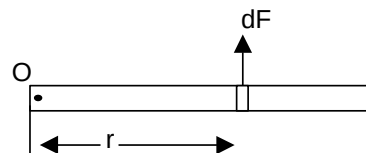
$$\Rightarrow \text{The total torque about } O = \tau = \int d\tau = \int r(iBdr)$$

$$\Rightarrow \tau = iB \int_0^L r dr = \frac{iBL^2}{2}$$

The angular acceleration $\alpha = \frac{\tau}{I}$

$$\Rightarrow \alpha = \left(\frac{iBL^2}{2} \right) / \left(\frac{mL^2}{3} \right)$$

$$\Rightarrow \alpha = \frac{3iB}{2m}$$



4. 2

Sol. By Gauss theorem, $\oint E dS = \frac{1}{\epsilon_0} \int dq$

$$\Rightarrow E \times 4\pi r^2 = \frac{4\pi \rho_0}{\epsilon_0} \int_0^r \left(1 - \frac{x}{3}\right) x^2 dx \Rightarrow E = \frac{\rho_0}{\epsilon_0} \left[\frac{r}{3} - \frac{r^2}{12} \right]$$

for max. electric field, $\frac{dE}{dr} = 0$

$$\Rightarrow \frac{1}{3} - \frac{r}{6} = 0 \Rightarrow r = \frac{6}{3} = 2$$

5. 5

Sol. From $P = \frac{V^2}{R}$

Resistance of heater,

$$R = \frac{V^2}{P} = \frac{(100)^2}{1000} = 10 \Omega$$

From, $P = i^2 R$

Current required across heater for power of 62.5 W

$$i = \sqrt{\frac{P}{R}} = \sqrt{\frac{62.5}{10}} = 2.5 A$$

Main current in the circuit,

$$I = \frac{100}{10 + \frac{10R}{10+R}} = \frac{10(10+R)}{10+2R}$$

This current will distribute in inverse ratio of resistance between heater and R.

$$\therefore i = \left(\frac{R}{10+R} \right) I$$

$$\text{or } 2.5 = \left(\frac{R}{10+R} \right) \left[\frac{10(10+R)}{10+2R} \right]$$

$$= \frac{10R}{10+2R}$$

Solving this equation, we get

SECTION - II: CHEMISTRY

PART - A

- | | | | |
|-------|-------|-------|-------|
| 1. C | 2. C | 3. C | 4. C |
| 5. A | 6. C | 7. B | 8. C |
| 9. A | 10. A | 11. A | 12. A |
| 13. B | 14. C | 15. B | 16. C |
| 17. C | 18. B | 19. D | 20. D |

PART - C

- | | | | |
|------|------|------|------|
| 1. 5 | 2. 3 | 3. 5 | 4. 9 |
| 5. 3 | | | |

SECTION- III: MATHEMATICS

PART – A

1. D

Sol. $-1 \leq \frac{1+x^4}{2x^2} \leq 1 \Rightarrow \frac{1+x^4}{2x^2} \leq 1$

$(\because \frac{1+x^4}{2x^2} \geq -1 \text{ is always true})$

$\Rightarrow (1-x^2)^2 \leq 0 \Rightarrow x = \pm 1$

2. B

Sol. (A) $f(4) = \frac{5}{2}$ and $f(-4) = \frac{-3}{2}$ Neither even nor odd

(B) $x - [x] = \text{fractional part of } x$

$0 \leq x - [x] < 1, [x - (x)] = 0$, constant hence even

(C) $f\left(\frac{\pi}{4}\right) = \left[\frac{2}{\sqrt{2}}\right] = 1; f\left(\frac{-\pi}{4}\right) = 0$

3. B

4. D

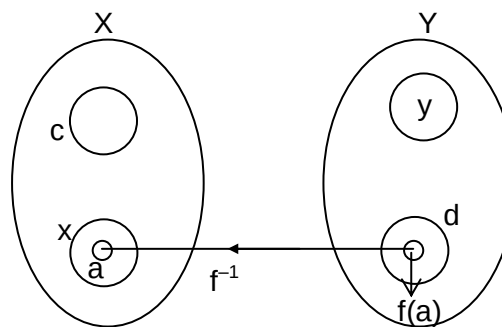
Sol. The given data is shown in the figure

Since $f^{-1}(d) = x$

$\Rightarrow f(x) = d$

Now, if $a \subset x$, $f(a) \subset d$

$\Rightarrow f^{-1}(f(a)) = a$.



5. C

Sol. $\int_{-2016}^{2016} \min(\{x\}, \{-x\}) dx$

$= 4032 \times 2 \times \int_0^{1/2} x dx = 4032 \times 2 \times \frac{1}{8} = 1008$

6. D

Sol. Put $x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$

$\Rightarrow I = \int_4^{1/4} t \sin\left(\frac{1}{t} - t\right) \cdot \left(-\frac{1}{t^2}\right) dt$

$= \int_4^{1/4} \frac{1}{t} \sin\left(t - \frac{1}{t}\right) dt = -I$

$\Rightarrow I = 0$

7. C

Sol. $e^{-x} f(x) = \left(2 + \int_0^x \sqrt{t^4 + 1} dt\right)$ if $x = 0$, $f(0) = 2$

$e^{-x} f'(x) - e^{-x} f(x) = \sqrt{x^4 + 1} \Rightarrow f'(0) = 3$

Let $g(x) = f^{-1}(x) \Rightarrow f(g(x)) = x \quad \because f(g(2)) = 2 \therefore g(2) = 0$

$f'(g(x)) \cdot g'(x) = 1 \Rightarrow f'(g(x)) \cdot g'(2) = 1$

$$\Rightarrow f'(0).g'(2)=1 \quad g'(2)=\frac{1}{3} \quad (f' \circ g)'(2)=\frac{1}{3}$$

8. D

Sol.

$$\begin{aligned} \text{Given limit} &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{\left(\frac{r}{n}\right)^3}{\left(\frac{r}{n}\right)^4 + 1} \quad \left(\because \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n f\left(\frac{r}{n}\right) = \int_0^1 f(x) dx \right) \\ &= \int_0^1 \frac{x^3}{x^4 + 1} dx = \frac{1}{4} (\log |1 + x^4|) \quad = \frac{1}{4} \log 2 \end{aligned}$$

9. B

Sol. If $S_n = \frac{1}{n^2 + 1} + \frac{2}{n^2 + 2} + \dots + \frac{n}{n^2 + n}$

$$T_1 + T_2 + \dots + T_n$$

$$T_r = \frac{r}{n^2 + r}$$

Now, if $r = 1, 2, \dots, n$

$$n^2 + 1 \leq n^2 + r \leq n^2 + n.$$

$$\Rightarrow \frac{1}{n^2 + n} \leq \frac{1}{n^2 + r} \leq \frac{1}{n^2 + 1} \Rightarrow \frac{r}{n^2 + n} \leq \frac{r}{n^2 + r} \leq \frac{r}{n^2 + 1}$$

$$\Rightarrow \frac{n(n+1)}{2(n^2 + n)} \leq S_n \leq \frac{n(n+1)}{2(n^2 + 1)} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{2} \leq \lim_{n \rightarrow \infty} S_n \leq \lim_{n \rightarrow \infty} \frac{n(n+1)}{2(n^2 + 1)} \Rightarrow \frac{1}{2} \leq \lim_{n \rightarrow \infty} S_n \leq \frac{1}{2} \Rightarrow \lim_{n \rightarrow \infty} S_n = \frac{1}{2}$$

10. A

Sol.
$$\lim_{n \rightarrow \infty} \frac{\left(1 + \frac{n+1}{n^2}\right)^{\frac{n^2}{n+1} \times \frac{n+1}{n}}}{\left(1 + \frac{2-3n}{n^2}\right)^{\frac{n^2}{2-3n} \times \frac{2-3n}{n}}} = \frac{e^1}{e^{-3}} = e^4.$$

11. C

12. D

13. A

Sol. $y = x^3 + ax^2 + bx + c$ has no extremum if y is either increasing or decreasing for all x

$$\frac{dy}{dx} = 3x^2 + 2ax + b > 0 \text{ for all } x$$

$$\text{If } (2a)^2 - 4 \times 3 \times b < 0 \Rightarrow a^2 < 3b$$

14. C

Sol. $f(0) \geq \lim_{x \rightarrow 0^+} f(x)$

$$\Rightarrow a \geq 1$$

15. B

Sol. For, $4x^2 + 9y^2 = 72$, $\frac{dy}{dx} = -\frac{4x}{9y} \Rightarrow \frac{dy}{dx} \Big|_{(3,2)} = -\frac{2}{3}$

$$\text{For } x^2 - y^2 = 5, \frac{dy}{dx} = \frac{x}{y} \Rightarrow \frac{dy}{dx} \Big|_{(3,2)} = \frac{3}{2}$$

$\therefore$ the curves cut orthogonally

16. B

Sol. If $g(x) = \frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx + e$

$$= \frac{1}{12} (3ax^4 + 4bx^3 + 6cx^2 + 12dx + 12e)$$

$$g(-4) = \frac{1}{12}(3.256a - 256 + 96c - 48d + 12e)$$

$$= \frac{1}{12}(16(48a - 16b + 6c - 3d) + 12e)$$

$$= e = g(0)$$

$$\Rightarrow g(x) \text{ satisfies Rolle's Theorem in } [-4, 0]$$

$$\Rightarrow g(x) \text{ has at least one root in } (-4, 0)$$

17. B

Sol. $\int \frac{dx}{4+5\sin x} = \int \frac{dt}{2t^2+5t+2} \left[\text{Put } \tan\left(\frac{x}{2}\right) = t \right]$

$$= \int \left[\frac{2}{3} \frac{1}{1+2t} - \frac{1}{3} \frac{1}{t+2} \right] dt$$

$$= \frac{1}{3} \ln \left| \frac{1+2t}{t+2} \right|$$

$$\Rightarrow p=3, q=2, r=2$$

$$\Rightarrow p-q+r=3.$$

18. B

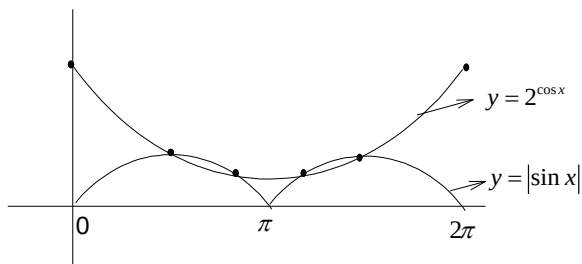
19. C

20. C

PART - C

1. 4

Sol.



2. 8

Sol. at $n = \infty$, f_n & f_{n-1} are same.

3. 3

Sol. $\int_0^3 f(x) dx = \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx$

$$= \int_0^1 f(x) dx + \int_0^1 f(x+1) dx + \int_0^1 f(x+2) dx$$

$$= \int_0^1 (3x^2 + 2x + 1) dx = \left[x^3 + x^2 + x \right]_0^1 = 3$$

4. 9

Sol. Points of discontinuity are $\frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{4}{3}, \frac{3}{2}, \frac{5}{3}, \frac{7}{3}, \frac{5}{2}, \frac{8}{3}$

5. 5

Sol. Because at $x = 1$, $2x + 3 = 5 \Rightarrow \lambda \leq 5$